WASP-52b

R-0532

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_240714` · created 2026-07-21T13:04:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460505.8284 +/- 0.0083 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.057
Transit depth (Rp/Rs)^2	3.68 +/- 2.17 [%]
Orbital Inclination (inc)	81.08 +/- 2.78
Airmass coefficient 1 (a1)	1.03 +/- 0.057
Airmass coefficient 2 (a2)	-0.0106 +/- 0.0078
Scatter in the residuals of the lightcurve fit is	5.59 %
Best Comparison Star	#3 - [348, 399]
Optimal Method	PSF photometry
Transit Duration (day)	0.03 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.057 vs 0.1646 (deviation 0.27σ)
Duration fit vs published	0.03 d vs 0.0758 d (deviation 0.83σ)
Mid-time O–C	-19.8 min
Depth SNR	1.9
Residual scatter	5.59 %

Light curve

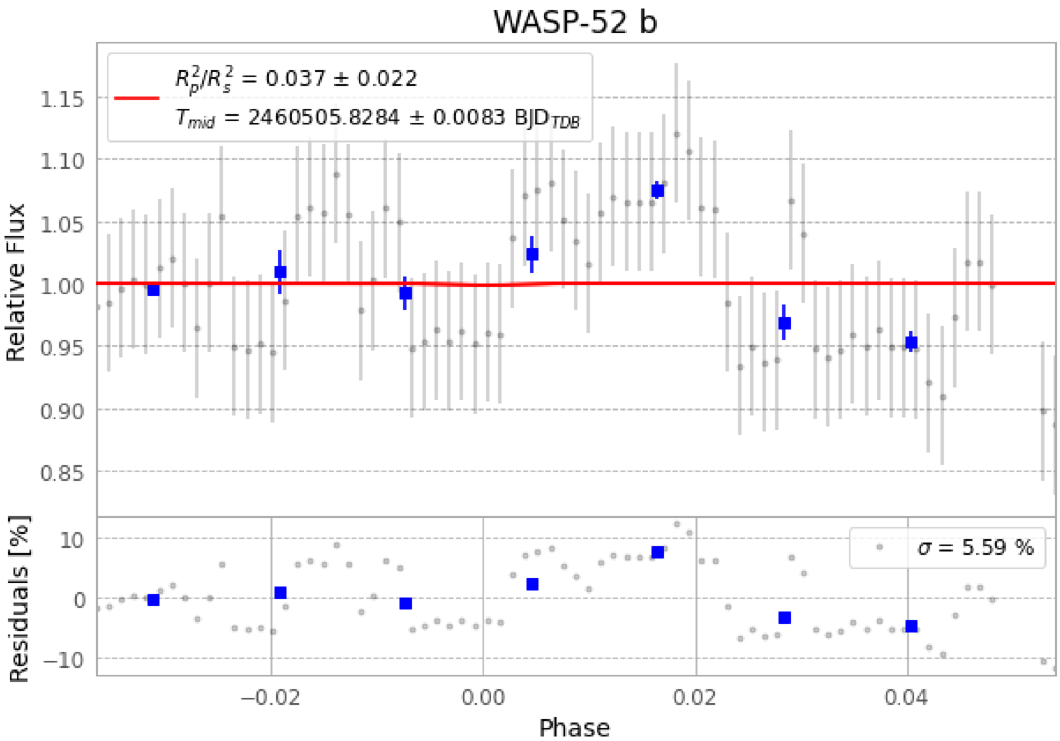

Figure 1 — FinalLightCurve_WASP-52 b_2024-07-13.png (SHA-256 92920061bdb3afd1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [519, 117], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 fda9ad6dc475d11c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

77 FITS frames (51 MB), `WASP-52240714061515.FITS` → `WASP-52240714100315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-240714.log (c02f1f08df1b…), FinalLightCurve_WASP-52 b_2024-07-13.png (92920061bdb3…), FinalLightCurve_WASP-52 b_2024-07-13.csv (2e5a122a3d19…)

Compute resources

Wall time 90.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)